London South Bank University

ENGINEERING PORTFOLIO / 2025 – 26

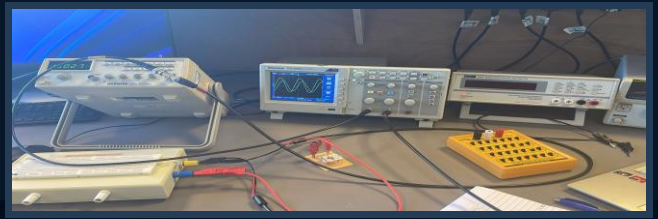
DC & AC
CIRCUIT ANALYSIS

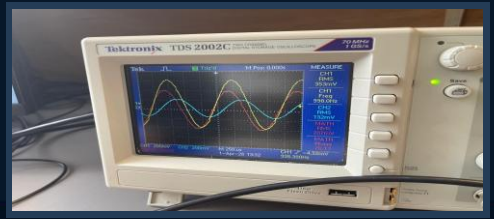
Hands-on investigation of electrical networks,
measurement and experimental validation

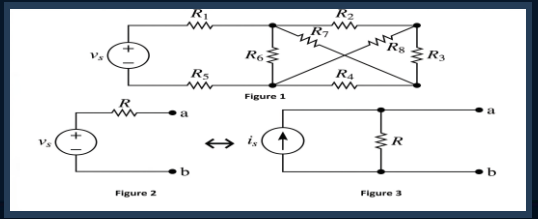


Electrical Circuit Analysis Laboratory | London South Bank University

From circuit theory to real laboratory measurements







BOKHTEAR UDDIN BADHON

BEng (Hons) Mechanical Engineering | First-Year Laboratory Portfolio

SEEKING A 2027 INDUSTRIAL PLACEMENT

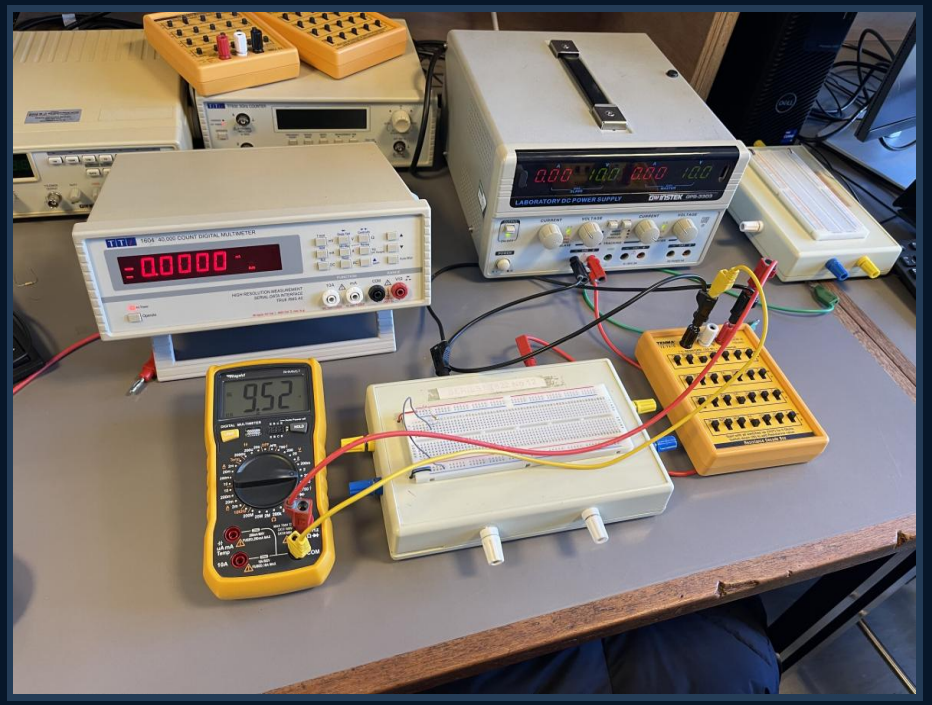
INNOVATIVE ENGINEERING.
A SUSTAINABLE TOMORROW.

PROJECT OVERVIEW

Five experiments. One engineering mindset.

2025–26 laboratory portfolio

- Build and test physical DC and AC circuits
- Measure voltage, current, resistance and waveform behaviour
- Compare theoretical predictions with experimental results
- Identify uncertainty, loading effects and measurement limitations



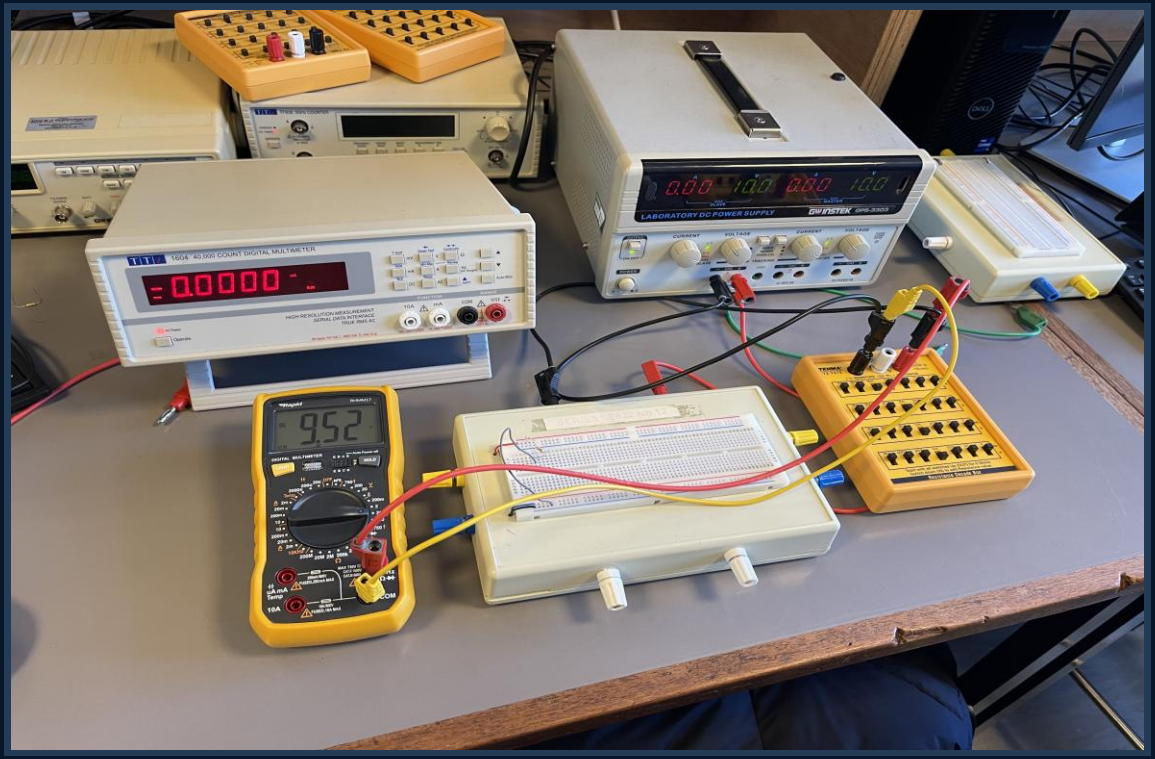
Original laboratory setup — DC resistance measurements

EXPERIMENTAL METHOD

Engineering workflow: from theory to evidence

01	**PLAN**	Select circuit model and expected behaviour
02	**BUILD**	Assemble components and check connections
03	**MEASURE**	Use DMM, power supply and oscilloscope
04	**ANALYSE**	Calculate, graph and compare with theory
05	**REFLECT**	Explain differences and limitations

Experiment 01 | Resistance and power



Breadboard, digital multimeters and DC power supply

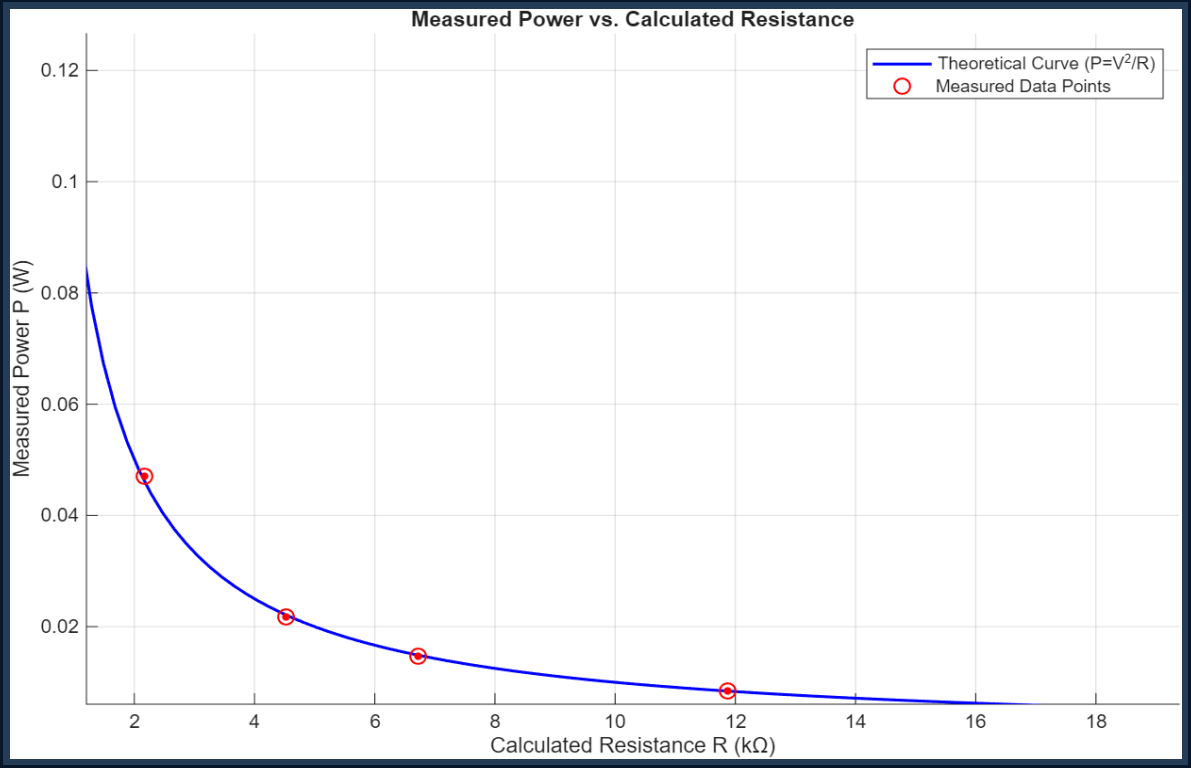
What I investigated

- Ohm’s law and power dissipation
- Resistor colour codes and ±5% tolerance
- Voltage and current divider behaviour
- Practical meter loading effects

10 V DC supply

1–18 kΩ resistors

Experiment 01 | Measured results



Original power-versus-resistance graph

Evidence from the logbook

- 1 kΩ nominal resistor measured at 0.979 kΩ (−2.13%)
- Voltage-divider deviations: 0.21–0.40%
- Measured power decreased as resistance increased
- All tested resistor values remained within ±5% tolerance

Engineering insight: accurate measurement requires understanding how the measuring instrument affects the circuit.

Experiment 02 | Kirchhoff’s laws



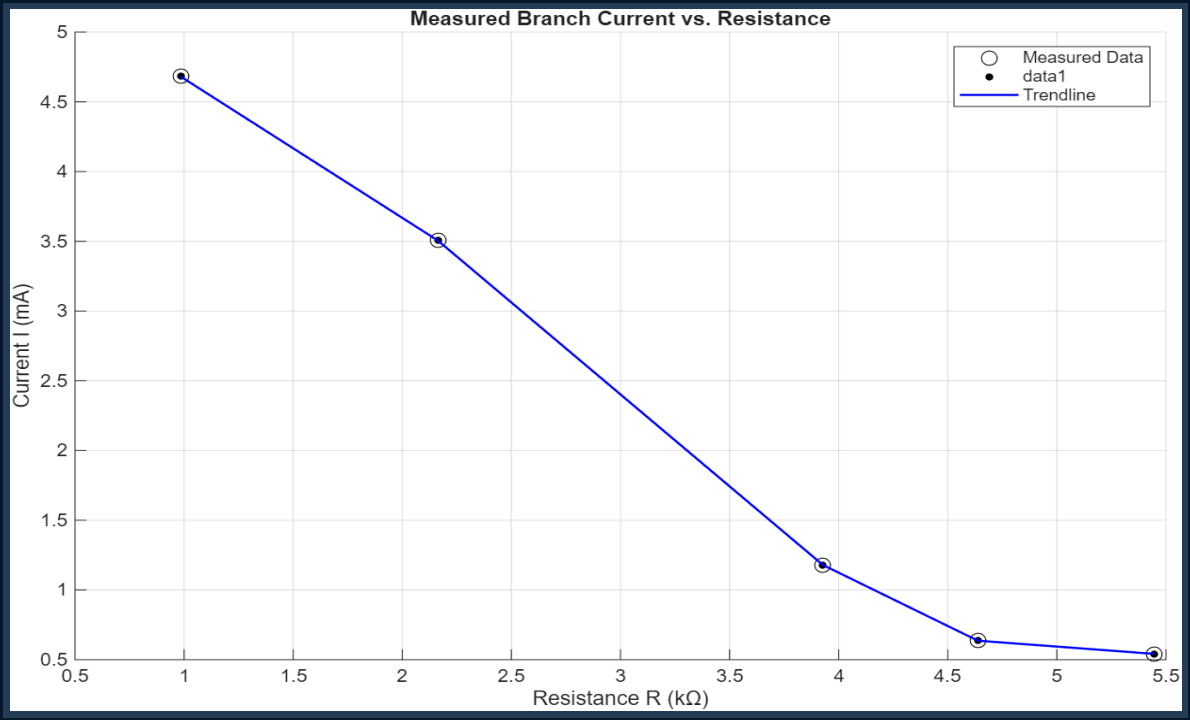
Original multi-resistor network and measurement setup

The engineering task

- Construct a resistive DC network
- Verify Kirchhoff’s voltage law across closed loops
- Verify current conservation at network nodes
- Use measured resistance values to refine predictions

12 V nominal DC source

Experiment 02 | Verification and error



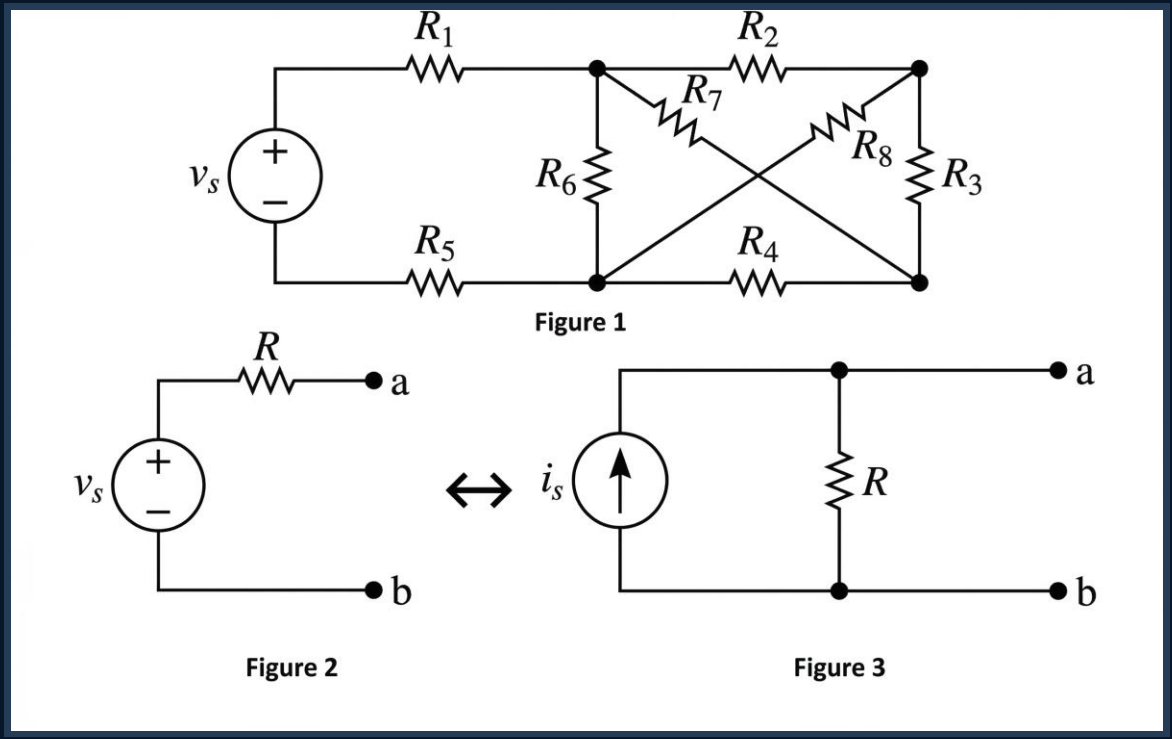
Measured branch current versus resistance

Reported observations

- Measured supply voltage: 12.190 V
- Loop-voltage residuals close to zero
- KCL agreement improved using measured resistances
- Tolerance, contact resistance and meter accuracy considered

Skill developed: using experimental evidence to test a theoretical network model.

Experiment 03 | Thévenin and Norton



Bridge network and its equivalent circuit representations

From complex to equivalent

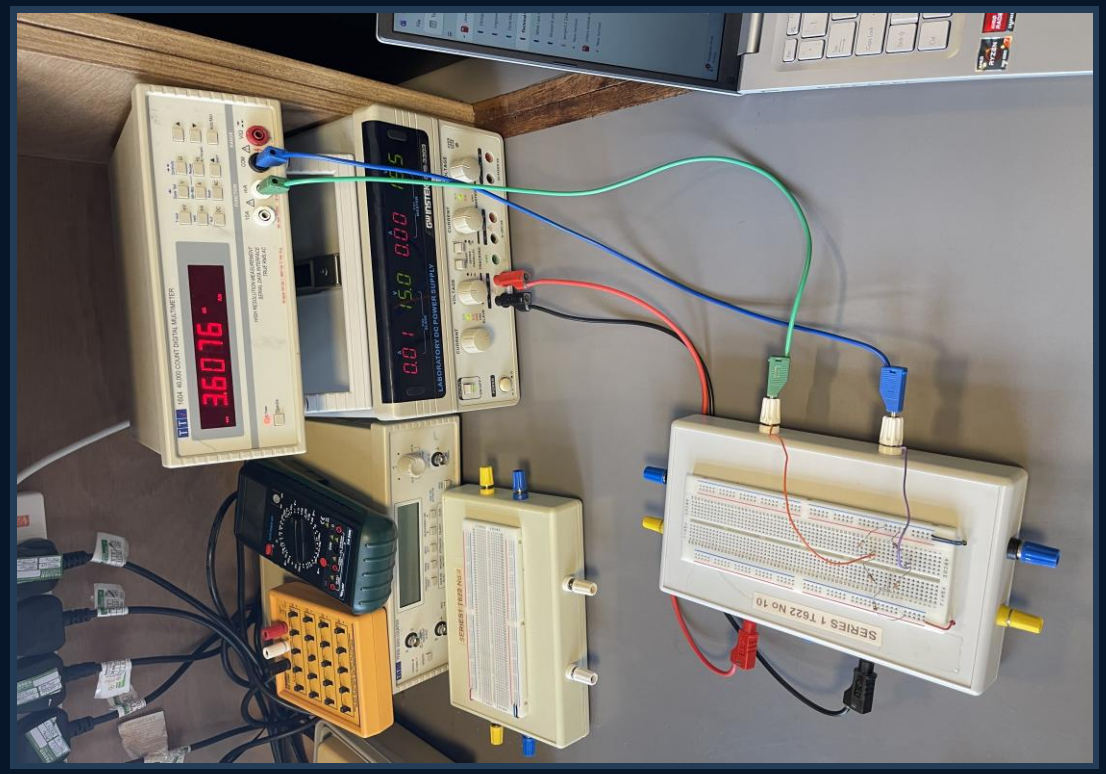
- Construct a multi-resistor bridge network
- Measure open-circuit voltage and short-circuit current
- Determine equivalent source parameters
- Predict load response and test the model

Vth = 10.720 V

In = 3.6076 mA

CRITICAL EVALUATION

Experiment 03 | When theory and reality differ



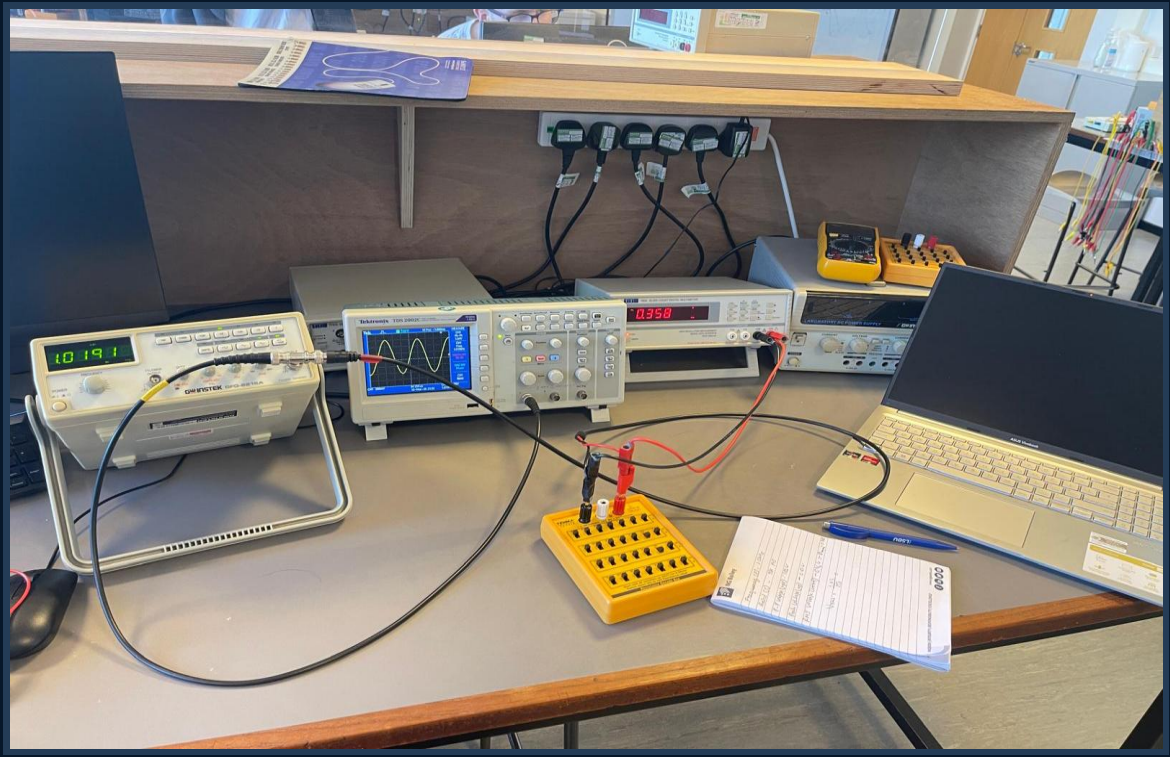
Bridge-network laboratory test

A result worth investigating

- The logbook reports 9.6% and 62.8% errors for two load comparisons
- Large deviations should not be hidden or treated as proof of accuracy
- Potential issues include load identification, wiring, reading and modelling
- Next step: repeat measurements and independently check the calculations

Engineering practice means questioning inconsistent results, not simply accepting them.

Experiment 04 | Oscilloscope fundamentals



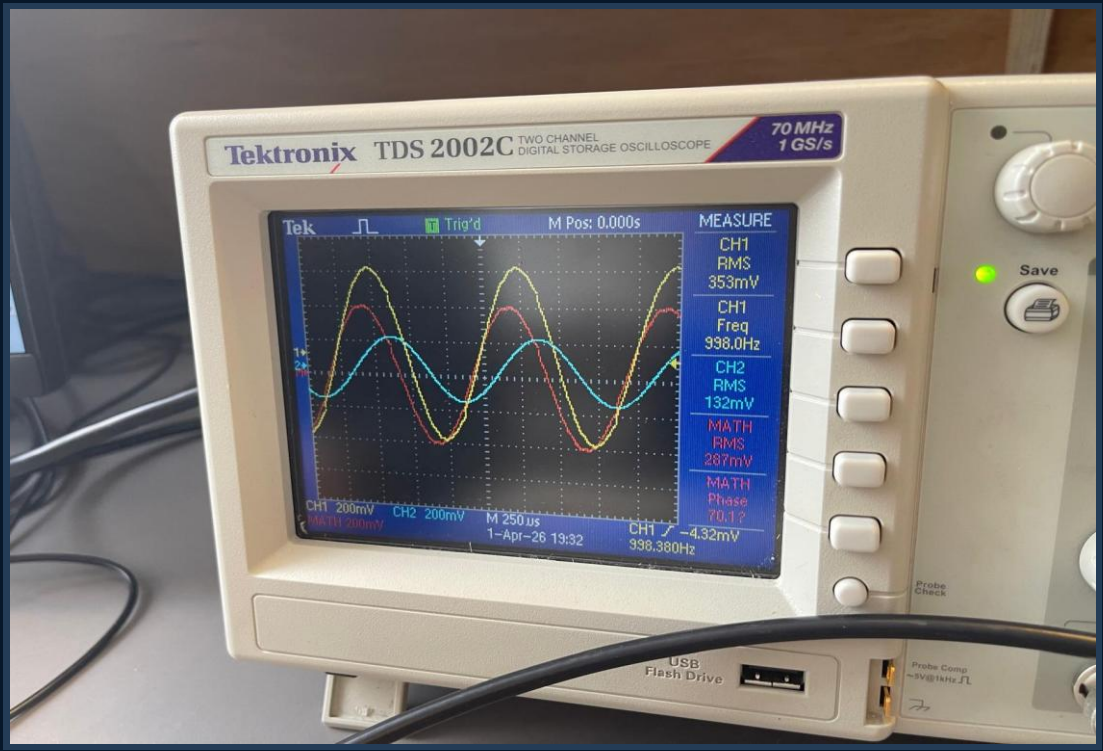
Function generator, oscilloscope and test circuit

Practical measurements

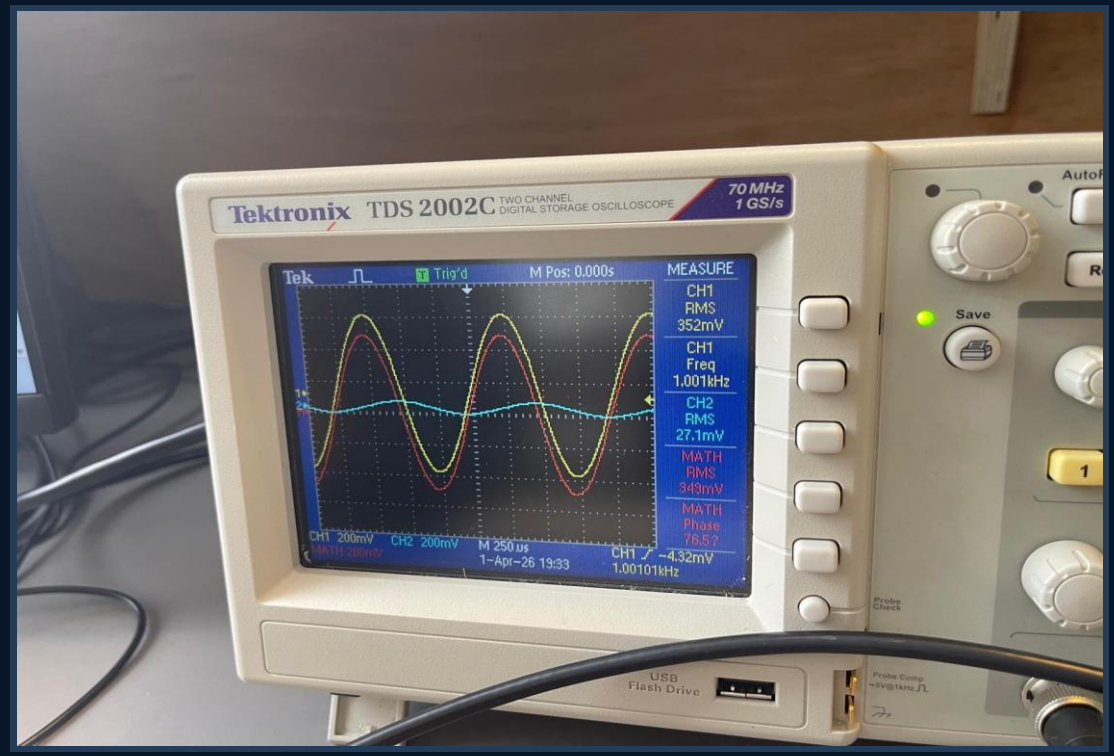
- Generate and stabilise a sinusoidal signal
- Measure peak-to-peak voltage and period
- Calculate frequency and RMS voltage
- Use oscilloscope triggering, cursors and channels

Generator ≈ 1.019 kHz

Experiment 04 | Voltage and phase



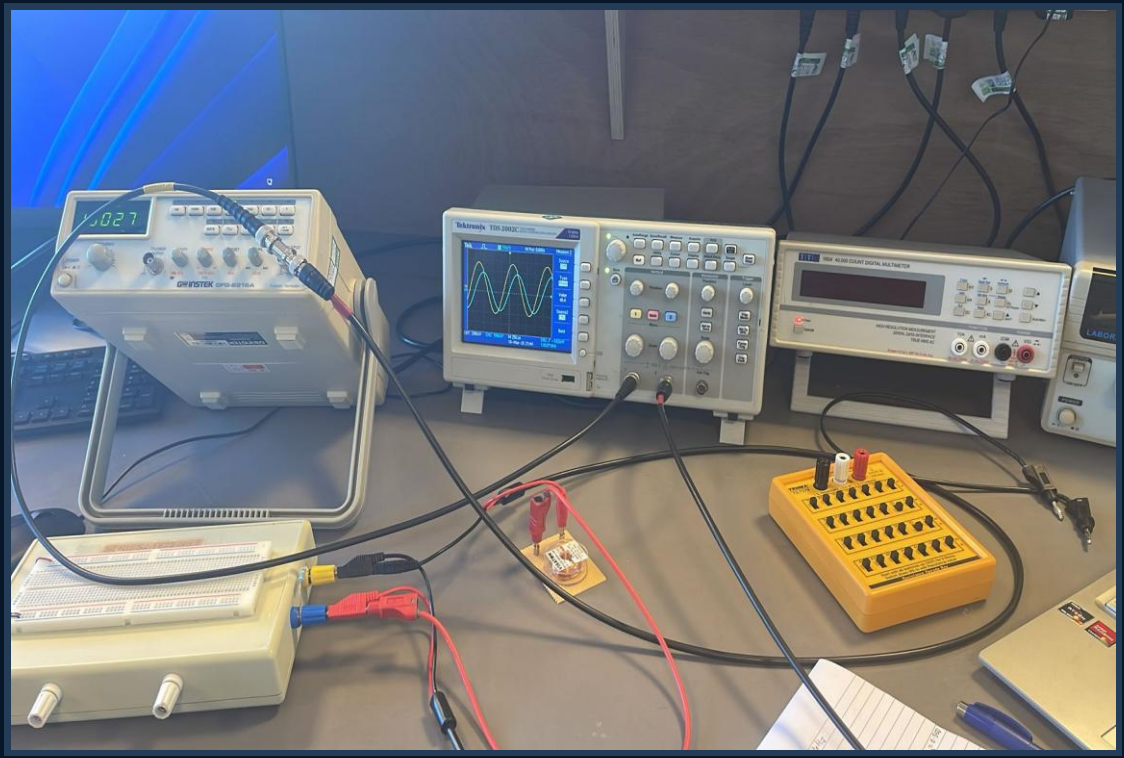
Original oscilloscope waveform photograph



Original two-channel waveform photograph

The two-channel method enables comparison of voltage and current timing in resistive and inductive loads.

Experiment 05 | Impedance and inductance

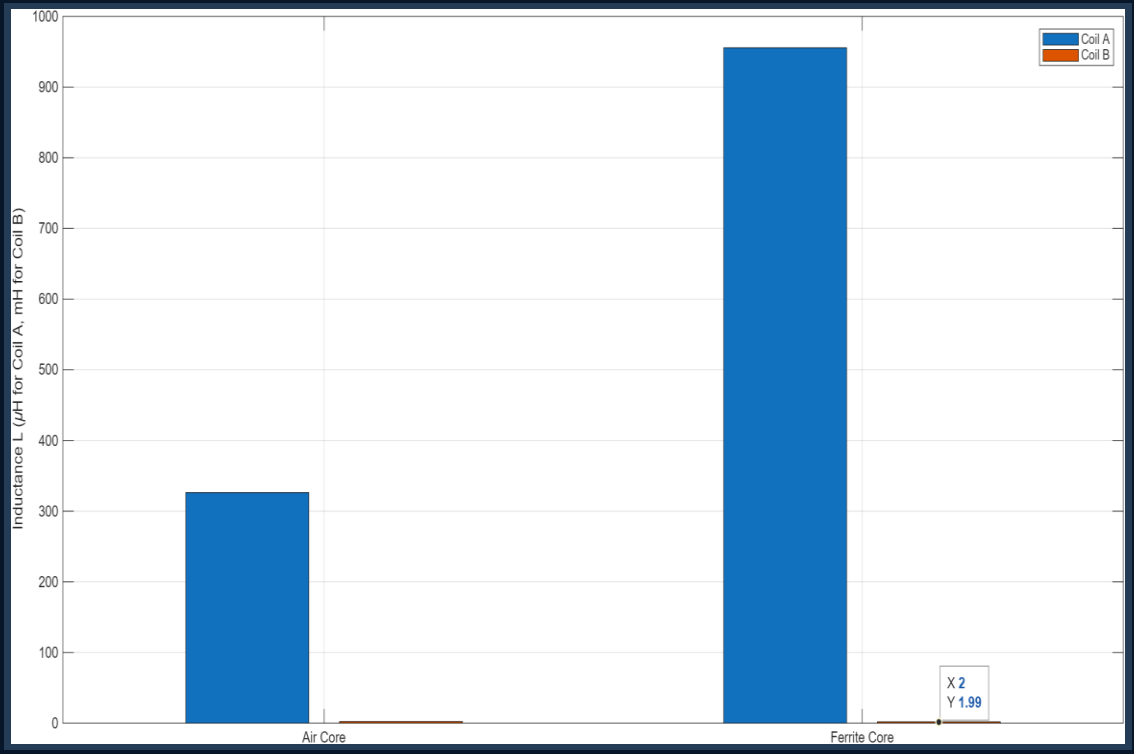


Original AC laboratory setup

Investigating coil behaviour

- Use a 1 Ω non-inductive shunt for current sensing
- Estimate impedance and phase angle from waveforms
- Compare air-core and ferrite-core coils
- Cross-check calculated inductance with an LCR meter

Experiment 05 | Ferrite-core effect



Original ferrite-core comparison graph

Coil A — reported values

326.3 μH

Air-core inductance

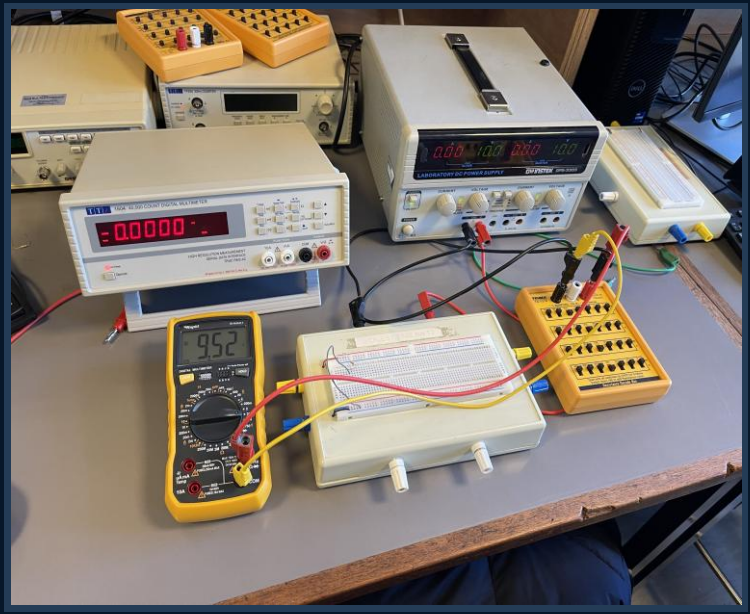
955.7 μH

With ferrite core inserted

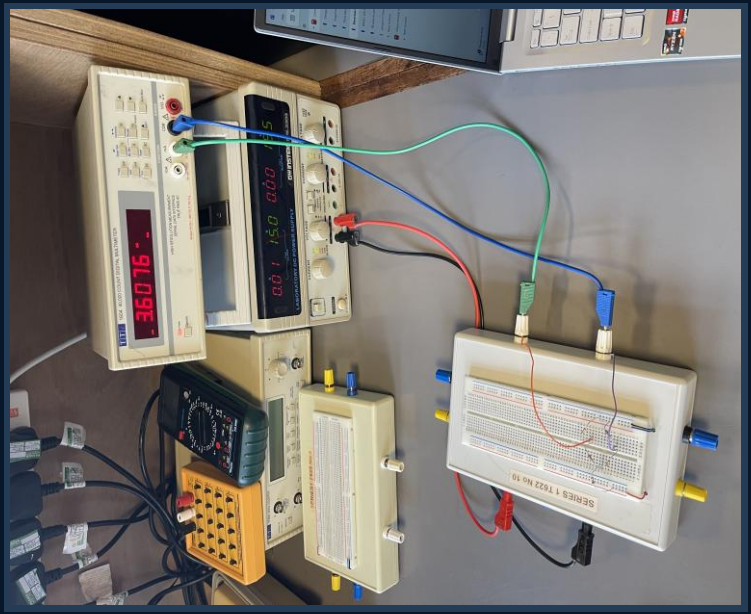
The higher-permeability core increases inductance and alters the phase relationship.

LABORATORY EQUIPMENT

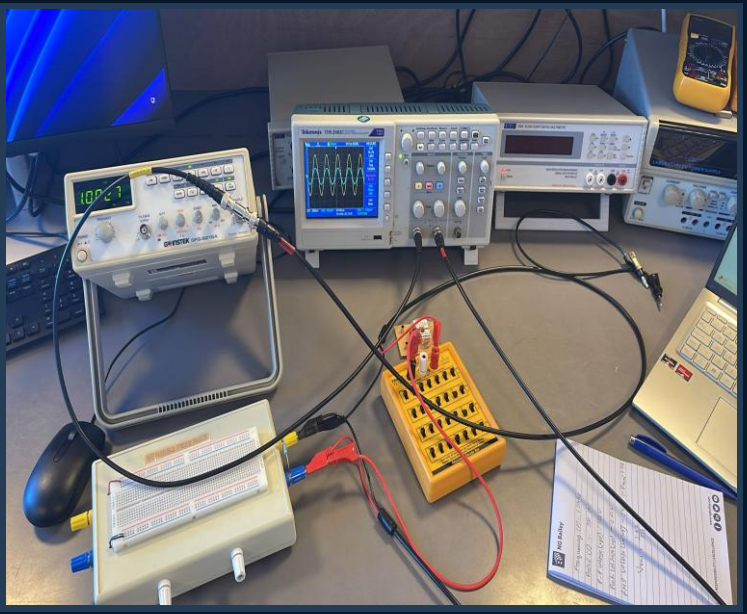
The tools behind the experiments



Digital multimeters and DC supply



Breadboard and resistive network



Oscilloscope and function generator

Equipment used: GW Instek DC supply • TTi 1604 DMM • Tektronix TDS 2002C oscilloscope • GW Instek function generator • breadboard • resistor/coil components • LCR meter

ENGINEERING COMPETENCIES

What the experimental work taught me

CIRCUIT ASSEMBLY	Breadboard construction, polarity and connections
INSTRUMENTATION	DMM, oscilloscope and function generator operation
ANALYTICAL REASONING	Ohm’s law, KCL/KVL, equivalent circuits and AC theory
DATA INTERPRETATION	Calculations, comparisons, charts and uncertainty
TECHNICAL COMMUNICATION	Structured laboratory reporting and visual evidence

Reflection | Developing as a mechanical engineer

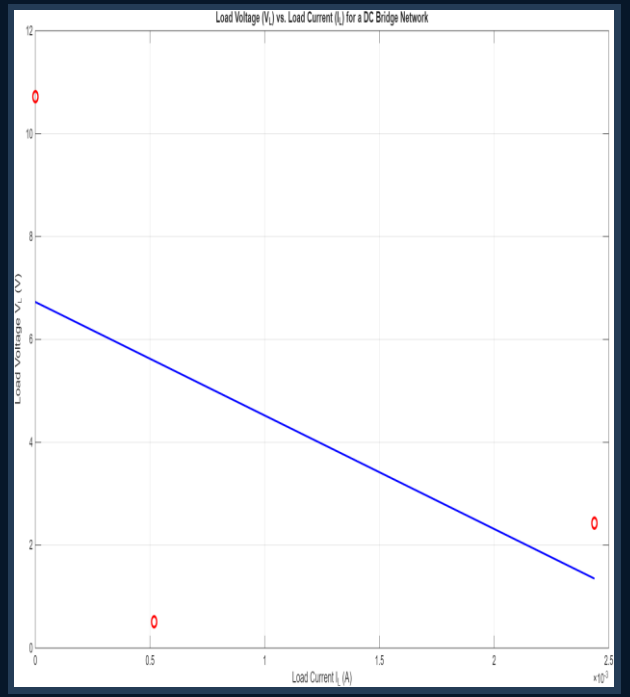
Why electrical analysis matters in mechanical engineering

- Modern mechanical systems integrate sensors, actuators, control hardware and electrical measurement.
- These experiments strengthened my confidence in testing physical systems and interpreting instrument data.
- Unexpected deviations reinforced the need to check assumptions, calculations and experimental setups.
- My next goal is to apply this foundation to mechatronics, testing, product development and industrial engineering.

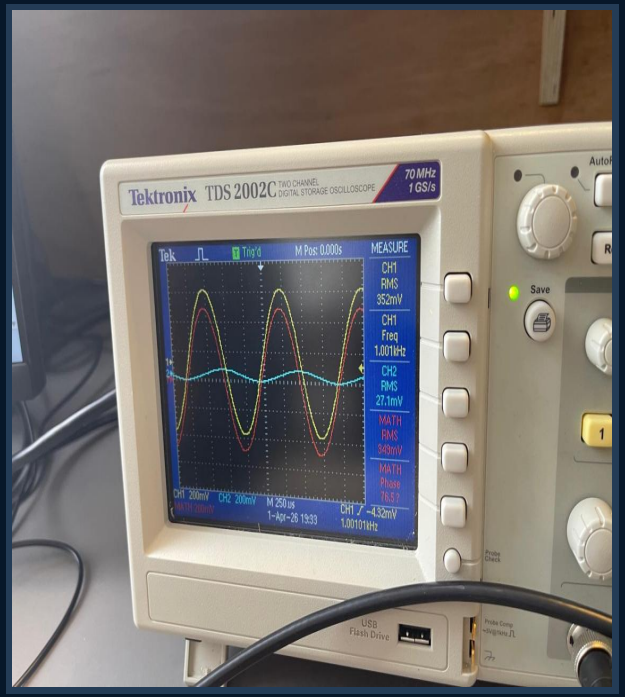
Evidence, documentation and next steps



Physical test evidence



Recorded V–I results



Oscilloscope evidence

Source: Electrical Circuit Analysis laboratory logbook, 2026. Figures and measurements reproduced from the student's own submitted document.

CONNECT & COLLABORATE

Seeking a 2027 industrial placement

BOKHTEAR UDDIN BADHON

BEng (Hons) Mechanical Engineering

London South Bank University

Interests: mechanical systems • testing • instrumentation • design • mechatronics

Open to engineering placement opportunities in 2027

